PLS-SEM Results

Respondents: 176 | Indicators: 29 | Bootstrap samples requested: 5000 (valid: 5000)

Model: X -> M -> Y, with a direct X -> Y path. Estimator: PLS-PM, Mode A, path weighting.

Confidence intervals: percentile bootstrap, 95%. P values are two-sided normal approximations. Seed: 42.

Indicators with loadings >= 0.708 meet the common guideline; lower loadings are flagged for review alongside AVE, reliability, and theory. These heuristics do not by themselves establish model validity.

01 Outer loadings

Construct	Indicator	Loading	Evaluation
X	X1	0.8232	Meets loading guideline
X	X2	0.7780	Meets loading guideline
X	X3	0.7941	Meets loading guideline
X	X4	0.8182	Meets loading guideline
X	X5	0.8319	Meets loading guideline
X	X6	0.7958	Meets loading guideline
X	X7	0.8324	Meets loading guideline
X	X8	0.8245	Meets loading guideline
X	X9	0.8373	Meets loading guideline
X	X10	0.8335	Meets loading guideline
X	X11	0.7104	Meets loading guideline
X	X12	0.7338	Meets loading guideline
X	X13	0.7961	Meets loading guideline
M	M1	0.8637	Meets loading guideline
M	M2	0.8318	Meets loading guideline
M	M3	0.8303	Meets loading guideline
M	M4	0.8515	Meets loading guideline
M	M5	0.8784	Meets loading guideline
M	M6	0.8409	Meets loading guideline
M	M7	0.8332	Meets loading guideline
M	M8	0.6172	Review with AVE, reliability, and theory
M	M9	0.8461	Meets loading guideline
Y	Y1	0.8440	Meets loading guideline

Construct	Indicator	Loading	Evaluation
Y	Y2	0.7799	Meets loading guideline
Y	Y3	0.8448	Meets loading guideline
Y	Y4	0.8497	Meets loading guideline
Y	Y5	0.8240	Meets loading guideline
Y	Y6	0.8680	Meets loading guideline
Y	Y7	0.8496	Meets loading guideline

02 Reliability AVE

Term	alpha	rhoA	rhoC	AVE
X	0.9534	0.9542	0.9589	0.6426
M	0.9398	0.9419	0.9500	0.6802
Y	0.9290	0.9322	0.9426	0.7015

03 HTMT

Term	X	M	Y
X			
M	0.5708		
Y	0.7693	0.6158	

04 Fornell Larcker

Term	X	M	Y
X	0.8016	0.5452	0.7280
M	0.5452	0.8248	0.5848
Y	0.7280	0.5848	0.8375

05 Cross loadings

Term	X	M	Y
X1	0.8232	0.4457	0.5544

Term	X	M	Y
X2	0.7780	0.4361	0.5592
X3	0.7941	0.4174	0.5458
X4	0.8182	0.4045	0.6186
X5	0.8319	0.4394	0.6237
X6	0.7958	0.3780	0.5899
X7	0.8324	0.4401	0.5465
X8	0.8245	0.4812	0.6384
X9	0.8373	0.4231	0.6338
X10	0.8335	0.4487	0.6302
X11	0.7104	0.5466	0.5081
X12	0.7338	0.3590	0.5435
X13	0.7961	0.4458	0.5747
M1	0.4486	0.8637	0.5318
M2	0.3809	0.8318	0.3681
M3	0.3916	0.8303	0.4284
M4	0.4874	0.8515	0.4592
M5	0.4809	0.8784	0.4987
M6	0.5269	0.8409	0.5082
M7	0.4311	0.8332	0.4894
M8	0.4181	0.6172	0.5103
M9	0.4366	0.8461	0.4977
Y1	0.6681	0.5576	0.8440
Y2	0.5300	0.3637	0.7799
Y3	0.5865	0.4189	0.8448
Y4	0.5893	0.5137	0.8497
Y5	0.5938	0.5223	0.8240
Y6	0.6215	0.5557	0.8680
Y7	0.6612	0.4674	0.8496

06 R square

Term	M	Y
R^2	0.2972	0.5803

Term	M	Y
AdjR^2	0.2932	0.5754

07 F square

Term	X	M	Y
X	0.0000	0.4229	0.5676
M	0.0000	0.0000	0.1197
Y	0.0000	0.0000	0.0000

08 Inner VIF

Predictor -> outcome	VIF
X -> M	1.0000
X -> Y	1.4229
M -> Y	1.4229

09 Direct effects

Effect	Original_O	Mean_M	STDEV	T_statistic	P_normal_approx	CI_2.5	CI_97.5
X -> M	0.5452	0.5526	0.0580	9.3986	0.0000	0.4332	0.6611
X -> Y	0.5822	0.5842	0.0779	7.4783	0.0000	0.4295	0.7324
M -> Y	0.2674	0.2687	0.0841	3.1797	0.0015	0.1001	0.4281

10 Indirect effects

Effect	Original_O	Mean_M	STDEV	T_statistic	P_normal_approx	CI_2.5	CI_97.5
X -> M -> Y	0.1458	0.1480	0.0482	3.0256	0.0025	0.0558	0.2463

11 Total effects

Effect	Original_O	Mean_M	STDEV	T_statistic	P_normal_approx	CI_2.5	CI_97.5
X -> Y	0.7280	0.7322	0.0442	16.4729	0.0000	0.6385	0.8116

12 Bootstrap loadings

Term	Original Est.	Bootstrap Mean	Bootstrap SD	T Stat.	2.5% CI	97.5% CI
X1 -> X	0.8232	0.8222	0.0306	26.8772	0.7556	0.8764
X2 -> X	0.7780	0.7776	0.0370	21.0359	0.6933	0.8386
X3 -> X	0.7941	0.7931	0.0296	26.8259	0.7290	0.8450
X4 -> X	0.8182	0.8180	0.0283	28.8744	0.7582	0.8678
X5 -> X	0.8319	0.8318	0.0249	33.4616	0.7793	0.8772
X6 -> X	0.7958	0.7948	0.0315	25.2435	0.7258	0.8501
X7 -> X	0.8324	0.8318	0.0299	27.8816	0.7661	0.8839
X8 -> X	0.8245	0.8239	0.0304	27.1045	0.7553	0.8758
X9 -> X	0.8373	0.8376	0.0256	32.6764	0.7817	0.8836
X10 -> X	0.8335	0.8328	0.0247	33.7233	0.7809	0.8777
X11 -> X	0.7104	0.7114	0.0495	14.3386	0.6055	0.7987
X12 -> X	0.7338	0.7330	0.0440	16.6745	0.6355	0.8082
X13 -> X	0.7961	0.7954	0.0346	22.9754	0.7169	0.8553
M1 -> M	0.8637	0.8628	0.0250	34.5389	0.8093	0.9062
M2 -> M	0.8318	0.8298	0.0304	27.3787	0.7648	0.8832
M3 -> M	0.8303	0.8295	0.0263	31.5306	0.7751	0.8746
M4 -> M	0.8515	0.8509	0.0219	38.8812	0.8036	0.8901
M5 -> M	0.8784	0.8785	0.0172	51.0187	0.8426	0.9092
M6 -> M	0.8409	0.8408	0.0220	38.1700	0.7936	0.8804
M7 -> M	0.8332	0.8323	0.0278	29.9221	0.7727	0.8809
M8 -> M	0.6172	0.6176	0.0586	10.5332	0.4955	0.7244
M9 -> M	0.8461	0.8451	0.0230	36.7871	0.7956	0.8866
Y1 -> Y	0.8440	0.8444	0.0255	33.0413	0.7897	0.8901
Y2 -> Y	0.7799	0.7785	0.0430	18.1229	0.6861	0.8528
Y3 -> Y	0.8448	0.8437	0.0239	35.3551	0.7916	0.8857
Y4 -> Y	0.8497	0.8495	0.0244	34.8518	0.7953	0.8917
Y5 -> Y	0.8240	0.8236	0.0333	24.7741	0.7512	0.8811
Y6 -> Y	0.8680	0.8674	0.0194	44.6478	0.8251	0.9016

Term	Original Est.	Bootstrap Mean	Bootstrap SD	T Stat.	2.5% CI	97.5% CI
Y7 -> Y	0.8496	0.8493	0.0221	38.4251	0.8030	0.8883

13 Bootstrap HTMT

Term	Original Est.	Bootstrap Mean	Bootstrap SD	T Stat.	2.5% CI	97.5% CI
X -> M	0.5708	0.5716	0.0625	9.1363	0.4432	0.6877
X -> Y	0.7693	0.7690	0.0477	16.1246	0.6686	0.8547
M -> Y	0.6158	0.6177	0.0618	9.9680	0.4910	0.7303